

KOWSALYA SARAVANAN

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Senior Software Engineer – AI, ML & Backend Development | GenAI Developer | MLOps Engineer
LinkedIn | Portfolio

EDUCATION

ASAN MEMORIAL COLLEGE

B.Sc. IN COMPUTER SCIENCE

2023 | Chennai, India

QUALITY THOUGHT

DATA SCIENCE CERT.

Specialized Training

SKILLS

BACKEND & LANGUAGES

Python | SQL | Bash
FastAPI | Flask | Asyncio
REST APIs | Microservices

GENAI / NLP

LangChain | LangGraph | LLMs
RAG Pipelines | Vector DBs
Prompt Engineering
Function Calling
Guardrails & Fallbacks

MLOPS, CLOUD & QA

Docker | Kubernetes (K8s) | CI/CD | AWS
Apache Airflow | Redis | Kafka
Unit Testing | Debugging
Logging & Observability
Local Model Deployment

COMPUTER VISION

InsightFace | YOLOv8 / YOLOv10
OpenCV | MediaPipe | SyncNet
Deepfake Detection
Object Detection

DATA SCIENCE & DL

Pandas | NumPy | Scikit-Learn
PyTorch | TensorFlow
Neo4j Graph DBs

SUMMARY

Senior Software Engineer specializing in backend development and AI/ML integrations with 2+ years of hands-on experience designing scalable microservices, REST APIs, and production-grade architectures. Experienced in LLM orchestration, prompt engineering, writing clean/testable code, and production monitoring.

EXPERIENCE

DATA SPARK AI SOLUTION | AI, ML & BACKEND ENGINEER (EDTECH)

June 2024 – Feb 2026 | Chennai, India

- **Infrastructure & Backend Services:** Orchestrated distributed microservices and recommendation workflows using **Airflow**, **Redis**, and local LLMs; established robust backend logging and performance monitoring.
- **Collaboration & Quality:** Participated in peer code reviews, promoted clean, testable software design, and utilized CI/CD pipelines to ensure stable, automated production releases.
- **Graph Recommenders:** Developed graph-based recommendation backend systems using **Neo4j** and **PySpark** to trace concept-level skill gaps.

ANUR CLOUD TECHNOLOGY | SOFTWARE ENGINEER (AI, ML & BACKEND)

Feb 2026 – June 2026 | Chennai, India

- **Clean Code & REST APIs:** Architected async, testable **REST APIs** using **FastAPI** to power backend services, ensuring sub-second inference latencies and writing thorough unit tests.
- **Secure FinTech Backends:** Engineered secure verification, lip-sync, and facial comparison backend APIs deployed for tier-1 FinTech banking systems.
- **System Reliability:** Maintained, troubleshot, and supported containerized applications using Docker and AWS, utilizing centralized logging and debugging practices.

PROJECTS

NEET, JEE & UPSC ADAPTIVE PLATFORM | EdTECH CORE - [SASTRA] DATA SPARK

- **Adaptive Orchestration Engine:** Architected a post-test remediation pipeline using **Airflow** and **Redis**. Designed robust system guardrails and fallback mechanisms to handle AI generation failures, routing requests to backup processes.
- **Interactive Classrooms:** Developed low-latency event-driven backend systems tracking live streams to trigger pop-up interactive quiz events.
- **AI Tutor & Translation:** Built a Whisper-based live translation microservice supporting selected languages and developed active subject-wise AI teachers.

LANGUAGES

English (Full Professional / C1)
Tamil (Native)

ACHIEVEMENTS

Pilot Member:

Selected as a founding pilot team backend engineer at Data Spark.

Banking Security:

Shipped secure production APIs for SBI, HDFC, and PNB.

Cost Savings:

Deployed offline local speech-to-text models to completely eliminate paid API overhead.

IELTS & PTE PREP PLATFORM | EdTECH CORE - DATA SPARK AI SOLUTION

- **Interactive Mock Bots:** Engineered specialized evaluation engines utilizing **Prompt Engineering**, **Function Calling**, and LLM integrations to deliver real-time, structured grading feedback.
- **CV Proctoring Engine:** Built a live camera monitoring backend system to perform eye-gaze tracking, triggering automated warnings and post-exam notification reports.

CBSE SCHOOL AUTOMATION SYSTEM | EdTECH INFRASTRUCTURE - DATA SPARK

- **Period-Wise Attendance:** Programmed automated computer-vision microservices using smart TV cameras to track student presence across classrooms.
- **Presence Monitoring:** Developed tracking algorithms verifying presence and mapping database entries to historical school attendance records.
- **SOS Alert System:** Deployed an integrated real-time emergency safety and anomaly alerting backend mechanism to flag safety hazards.

CONVERSATIONAL INTERVIEW & ESL APP | EdTECH TOOLING - DATA SPARK

- **Mock Interview Practice:** Developed video-based mock interview practice bots evaluating candidates' visual delivery, eye contact, and response timing.
- **ESL Speaking Bot:** Built interactive English learning bots supporting natural voice conversations with automatic corrections.

FINTECH BIOMETRIC & SPEECH SUITE | FinTECH CORE - ANUR CLOUD TECHNOLOGY

- **Local STT Transformation:** Successfully replaced cloud-dependent Google speech API keys with self-hosted, offline multilingual STT models inside on-premises environments.
- **Secure Banking APIs:** Shipped clean, testable backend verification REST APIs (face comparison, liveness check via **MiniFASNetV2**) for **SBI**, **HDFC**, and **PNB**.
- **Deepfake Detection:** Integrated **SyncNet** to run audio-visual synchronization checks and identify anomalies relative to speech streams.
- **FFmpeg Pipeline Optimization:** Programmed custom backend video compression microservices with FFmpeg, shrinking client-side transfer overhead by 90%.